Linear Algebra FINAL

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [8 pts] Solve 6x+4y=14 and 4x+6y=16. Show elimination and verify by substitution.

2. [6 pts] For A=[[2,2],[2,6]], compute det(A) and state whether A is invertible. Justify the conclusion.

3. [8 pts] Let T(x,y)=(x+2y, 3x–y). Find T(1,–1), write the matrix of T, and state whether T is linear.

4. [6 pts] Give a test for linear independence of vectors and apply it to {(1,0),(0,1)}.

5. [5 pts] Explain geometrically what the null space of a matrix represents and how it relates to Ax=0.

6. [8 pts] Solve 6x+1y=8 and 4x+6y=16. Show elimination and verify by substitution.

7. [6 pts] For A=[[4,2],[2,4]], compute det(A) and state whether A is invertible. Justify the conclusion.

8. [8 pts] Let T(x,y)=(x+2y, 3x–y). Find T(1,–1), write the matrix of T, and state whether T is linear.

9. [6 pts] Give a test for linear independence of vectors and apply it to {(1,0),(0,1)}.

10. [5 pts] Explain geometrically what the null space of a matrix represents and how it relates to Ax=0.